

SUMMER 2014

Problem 1. (3 Points)

Let $z_1 = r_1 e^{j\theta_1}$ and $z_2 = r_2 e^{j\theta_2}$. Find $z = z_1 \cdot z_2$ in the form of $z = a + jb$ where $r_1, \theta_1, r_2, \theta_2, a, b \in \mathbb{R}$.

Note that $e^{j\theta_1} = \cos(\theta_1) + j \sin(\theta_1)$

$$z = z_1 \cdot z_2 = r_1 r_2 e^{j(\theta_1 + \theta_2)} = r_1 r_2 [\cos(\theta_1 + \theta_2) + j \sin(\theta_1 + \theta_2)] = a + jb$$

$$a = r_1 r_2 \cos(\theta_1 + \theta_2), \quad b = r_1 r_2 \sin(\theta_1 + \theta_2)$$

Problem 2. (3 Points)

Let $z_1 = 1 + 2j$ and $z_2 = 2 + j$.

a. Find $z_1 \cdot z_2 = 5j$

b. Find $\frac{z_1}{z_2} = \frac{4}{5} + j\frac{3}{5}$

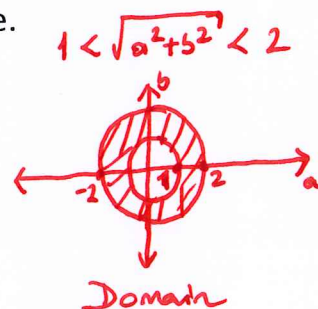
c. Find $z_1 + z_2 = 3 + 3j$

d. Find $z_1 - j z_2 = 2$

Problem 3. (4 Points)

Let $w(z) = j \cdot (z - 2)$ is a mapping function on the domain $\{z \in \mathbb{C}, 1 < |z| < 2\}$. Find the range of the mapping function $w(z)$ for the given domain. Roughly sketch domain and the range in the complex plane.

$$* \quad z = a + jb \Rightarrow |z| = \sqrt{a^2 + b^2} \Rightarrow$$



$$* \quad w(z) = u + jv = j(a + jb - 2) \\ = -b + j(a - 2)$$

$$\underline{u = -b}, \quad \underline{v = a - 2}$$

$$\Rightarrow \begin{aligned} 1 &< \sqrt{a^2 + b^2} < 2 \\ 1 &< \sqrt{(v+2)^2 + u^2} < 2 \end{aligned}$$

